

ECO 340 PROJECT DRAFT 2

NITYA TANDON

```
rm(list = ls())
```

```
options(scipen = 20)
```

```
if (!require("jtools")) install.packages("jtools")
```

```
library(jtools)
```

```
if (!require("forcats")) install.packages("forcats")
```

```
library(forcats)
```

```
if (!require("tidyr")) install.packages("tidyr")
```

```
library(tidyr)
```

```
if (!require("dplyr")) install.packages("dplyr")
```

```
library(dplyr)
```

```
if (!require("expss")) install.packages("expss")
```

```
library(expss)
```

```
if (!require("stringr")) install.packages("stringr")
```

```
library(stringr)
```

```
if (!require("ggplot2")) install.packages("ggplot2")
```

```
library(ggplot2)
```

```
if (!require("sandwich")) install.packages("sandwich")
```

```
library(sandwich)
```

```

if (!require("lmtest")) install.packages("lmtest")

library(lmtest)

if (!require("huxtable")) install.packages("huxtable")

library(huxtable)

if (!require("AER")) install.packages("AER")

library(AER)

if (!require("plm")) install.packages("plm")

library(plm)

if (!require("car")) install.packages("car")

library(car)

if (!require("stargazer")) install.packages("stargazer")

library(stargazer)

if (!require("clubSandwich")) install.packages("clubSandwich")

library(clubSandwich)

if (!require("broom")) install.packages("broom")

library(broom)

```

LOADING AND CLEANING DATA:

```

load("C:/Users/nitya/OneDrive/Desktop/ECO 340 FINAL PROJECT/nlsy97_long.Rda")

names(nlsy97_long)

```

```

[1] "personid"      "year"          "marstat"
[4] "citizenship"   "highestgrade"  "jhgrades"

```

[7]	"hsgrades"	"highested1"	"highested2"
[10]	"highested3"	"union"	"fsize1"
[13]	"fsize2"	"hadinc"	"workinc1"
[16]	"workinc2"	"workinc_est"	"student1"
[19]	"student2"	"tenure"	"emphours"
[22]	"tothours"	"sex"	"birthmn"
[25]	"birthyr"	"samplotype"	"b_dad_ed"
[28]	"b_mom_ed"	"r_dad_ed"	"r_mom_ed"
[31]	"race"	"asvab"	"organized_02"
[34]	"conscientious_02"	"undependenable_02"	"thorough_02"
[37]	"agreeable_02"	"difficult_02"	"stubborn_02"
[40]	"trustful_02"	"extrovert_08"	"critical_08"
[43]	"dependable_08"	"anxious_08"	"open_08"
[46]	"reserved_08"	"sympathetic_08"	"disorganized_08"
[49]	"calm_08"	"conventional_08"	"hardworker_08"
[52]	"amountwork_08"	"highstandards_08"	"effortwork_08"
[55]	"rulebreaker_08"	"schoolrebel_08"	"traditional_08"
[58]	"rulefollower_08"	"hardworker_10"	"amountwork_10"
[61]	"highstandards_10"	"effortwork_10"	"rulebreaker_10"
[64]	"schoolrebel_10"	"traditional_10"	"rulefollower_10"
[67]	"teenempjobs"	"adulttempjobs"	"adulttotjobs"

Making Birth year and finding out age

```
nlsy97_long$birthyr <- as.numeric(nlsy97_long$birthyr)
nlsy97_long$year <- as.numeric(nlsy97_long$year)
nlsy97_long$age <- nlsy97_long$year - nlsy97_long$birthyr

summary(nlsy97_long$age)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
11.00	20.00	27.00	27.49	35.00	44.00

Creating the education variable and cleaning it and filled up

```
nlsy97_long <- nlsy97_long %>% mutate(highested= case_when(
  year == 1997 ~ highestgrade,
  year >= 1998 & year <2010 ~ highested1,
  year >= 2010 ~ highested2))

summary(nlsy97_long$highested)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
0.00	10.00	12.00	12.49	14.00	95.00	123380

```
nlsy97_long <- nlsy97_long %>% mutate(highested= case_when(
  highested == 0 ~ NA,
  highested == 95 ~ NA,
  TRUE ~ highested))

summary(nlsy97_long$highested)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
1.00	10.00	12.00	12.39	14.00	20.00	123582

```
nlsy97_long <- nlsy97_long %>%
  arrange(personid, year) %>%
  group_by(personid) %>%
  fill(highested, .direction = "down") %>%
  ungroup()

summary(nlsy97_long$highested)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
1.00	11.00	12.00	12.65	15.00	20.00	18031

Taking out missing variables

```
nlsy97_cleaned <- nlsy97_long %>%
  drop_na(highested) %>%
  filter(year != 2012 & year != 2014 & year != 2016 &
         year != 2018 & year != 2020 & year != 2022 & year != 2024)
```

Table

```
nlsy97_long %>% count(year)
```

```
nlsy97_cleaned %>% count(year)
```

```
aggregate(highested ~ age, data = nlsy97_cleaned, summary)
```

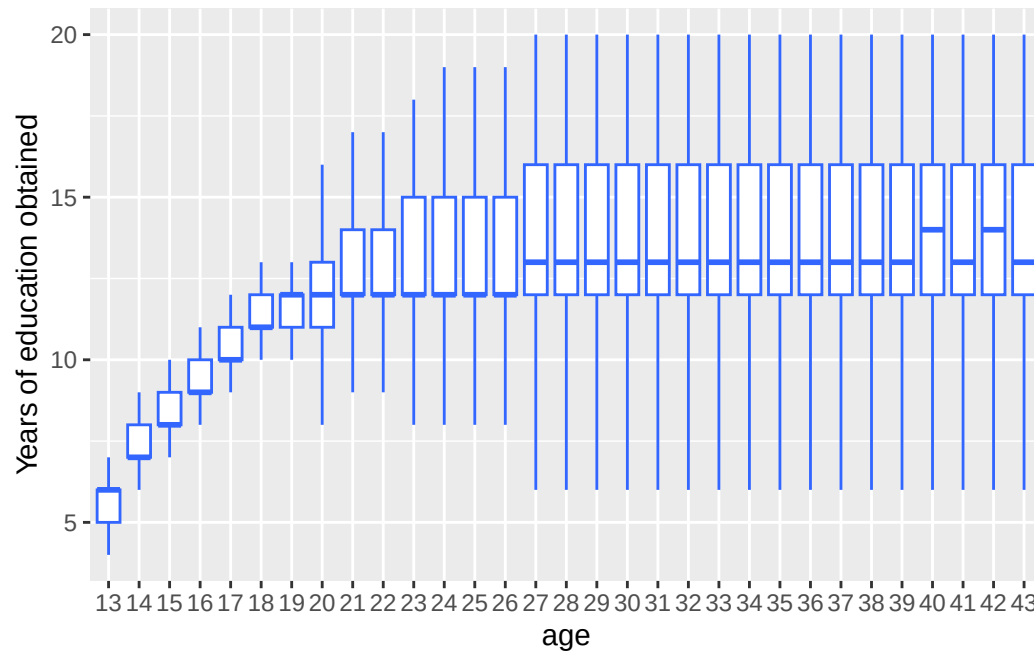
```
cor(nlsy97_cleaned$age, nlsy97_cleaned$highested)
```

```
[1] 0.4509697
```

Box plot

```
plot1 <- ggplot(nlsy97_cleaned, aes(as.factor(age), highested)) +
  geom_boxplot(outliers = FALSE, fill = "white", colour = "#3366FF") +
  xlab("age") +
  ylab("Years of education obtained")

print(plot1)
```



CREATING VARIABLES FOR WAGE AND INCOME AS WELL AS CONTROL VARIABLES

Focusing on worker and hours worked, etc, with only looking at employed workers with income and then creating a work income. Then calculating work income, wage and log wage

```
nlsy97_cleaned <- nlsy97_cleaned %>% select(year, personid, workinc2, emphours, hadinc, workinc)

nlsy97_cleaned <- subset(nlsy97_cleaned, hadinc==1 & !is.na(emphours) & emphours>0)

nlsy97_cleaned <- nlsy97_cleaned %>% mutate(workinc= case_when(
  workinc_est==1 ~ 2500,
```

```

workinc_est==2 ~ 7500,

workinc_est==3 ~ 17500,

workinc_est==4 ~ 37500,

workinc_est==5 ~ 75000,

workinc_est==6 ~ 150000,

workinc_est==7 ~ 300000,

TRUE ~ workinc2))

nlsy97_cleaned$wage <- nlsy97_cleaned$workinc/nlsy97_cleaned$emphours

nlsy97_cleaned <- nlsy97_cleaned %>% select(year, personid, wage, workinc, emphours, everything())

nlsy97_cleaned <- subset(nlsy97_cleaned, wage>0 & !is.na(wage))

nlsy97_cleaned$lnwage <- log(nlsy97_cleaned$wage)

nlsy97_cleaned <- nlsy97_cleaned %>% select(year, personid, lnwage, everything())

summary(nlsy97_cleaned$workinc)

```

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	6000	18000	27849	37500	416374	

```
summary(nlsy97_cleaned$wage)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0014	5.1948	11.1313	21.5369	20.1223	30000.0000

```
summary(nlsy97_cleaned$lnwage)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-6.551	1.648	2.410	2.276	3.002	10.309

control variables = age square and age cube

```
nlsy97_cleaned$agesq <- nlsy97_cleaned$age^2  
nlsy97_cleaned$agecu <- nlsy97_cleaned$age^3
```

```
summary(nlsy97_cleaned$agesq)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
169.0	441.0	676.0	760.5	961.0	1849.0

```
summary(nlsy97_cleaned$agecu)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2197	9261	17576	23097	29791	79507

Control variables = calculating tenure

```
nlsy97_cleaned$tenure <- nlsy97_cleaned$tenure / 52  
summary(nlsy97_cleaned$tenure)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
0.0000	0.5385	1.5961	2.9285	3.8269	31.5769	2015

Control variable = Female variable

```
nlsy97_cleaned$female <- as.numeric(nlsy97_cleaned$sex) - 1  
summary(nlsy97_cleaned$female)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.000	0.000	0.495	1.000	1.000

Control ASVAB

```
nlsy97_cleaned$asvab <- as.numeric(nlsy97_cleaned$asvab) / 1000  
summary(nlsy97_cleaned$asvab)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
0.00	23.82	48.31	49.23	74.28	100.00	17394

Non student age restriction

```
nlsy97_cleaned <- nlsy97_cleaned %>% mutate(student= case_when(  
  year == 1997 ~ student2,  
  year >= 1998 & year <2005 ~ student1,  
  year >= 2005 ~ student2))  
  
nlsy_agerestric <- subset(nlsy97_cleaned, student < 8 & age >= 30)  
  
nlsy_agerestric$experience_1 <- nlsy_agerestric$age - nlsy_agerestric$highested - 6  
  
summary(nlsy_agerestric$experience_1)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
4.00	12.00	15.00	15.45	19.00	31.00

REGRESSION 1: the baseline regression

```
reg1 <- lm(  
  lnwage ~ highested + age + agesq + agecu + union + tenure + female + as.factor(year),  
  data = nlsy_agerestric  
)  
  
summary(reg1)
```

Call:

```
lm(formula = lnwage ~ highested + age + agesq + agecu + union +  
    tenure + female + as.factor(year), data = nlsy_agerestric)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.4209	-0.3559	0.0174	0.3790	7.1105

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-6.8688954	6.3371795	-1.084	0.27842
highested	0.1014897	0.0016544	61.345	< 0.0000000000000002 ***
age	0.6267746	0.5294040	1.184	0.23645
agesq	-0.0156978	0.0146220	-1.074	0.28302
agecu	0.0001308	0.0001337	0.978	0.32786
union	0.0254450	0.0135134	1.883	0.05972 .
tenure	0.0256116	0.0010335	24.781	< 0.0000000000000002 ***
female	-0.2354071	0.0099305	-23.705	< 0.0000000000000002 ***
as.factor(year)2011	-0.0647954	0.0348077	-1.862	0.06268 .
as.factor(year)2013	-0.0785909	0.0347411	-2.262	0.02369 *
as.factor(year)2015	-0.0623024	0.0391447	-1.592	0.11149
as.factor(year)2017	0.0844657	0.0438743	1.925	0.05422 .
as.factor(year)2019	0.0799781	0.0467745	1.710	0.08730 .

```
as.factor(year)2021  0.1539066  0.0497655  3.093 0.00199 **
as.factor(year)2023  0.4558711  0.0528176  8.631 < 0.0000000000000002 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7743 on 24995 degrees of freedom

(3521 observations deleted due to missingness)

Multiple R-squared: 0.2262, Adjusted R-squared: 0.2258

F-statistic: 521.9 on 14 and 24995 DF, p-value: < 0.00000000000000022

```
reg1a <- coeftest(reg1, vcov = vcovHC, type = "HCO")
reg1b <- coeftest(reg1, vcov = vcovCL, type = "HCO", cluster = ~personid)

export_summs(
  reg1, reg1a, reg1b,
  model.names = c("Basic", "Robust SE", "Clustered SE"),
  digits = 4,
  coefs = c("highested", "age", "agesq", "agecu", "union", "tenure", "female")
)
```

REGRESSION 2: the baseline regression with asvab as a control

```
reg2 <- lm(
  lnwage ~ highested + asvab + age + agesq + agecu + union + tenure + female + as.factor(year)
  data = nlsy_agerestric
)

summary(reg2)
```

Call:

```
lm(formula = lnwage ~ highested + asvab + age + agesq + agecu +
    union + tenure + female + as.factor(year), data = nlsy_agerestric)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.4919	-0.3498	0.0169	0.3673	6.8993

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.19797748	6.82675123	-0.468	0.63947
highested	0.06650312	0.00219401	30.311	< 0.0000000000000002 ***
asvab	0.00625233	0.00022135	28.246	< 0.0000000000000002 ***
age	0.35541155	0.57032901	0.623	0.53318
agesq	-0.00879509	0.01575330	-0.558	0.57664
agecu	0.00007354	0.00014408	0.510	0.60974

union	0.06499327	0.01478430	4.396	0.0000111 ***
tenure	0.02436439	0.00111684	21.815	< 0.0000000000000002 ***
female	-0.23123678	0.01073328	-21.544	< 0.0000000000000002 ***
as.factor(year)2011	-0.06417966	0.03808536	-1.685	0.09197 .
as.factor(year)2013	-0.06725220	0.03792036	-1.774	0.07616 .
as.factor(year)2015	-0.02989843	0.04263048	-0.701	0.48310
as.factor(year)2017	0.11476795	0.04767447	2.407	0.01608 *
as.factor(year)2019	0.10340703	0.05078742	2.036	0.04176 *
as.factor(year)2021	0.17723515	0.05395628	3.285	0.00102 **
as.factor(year)2023	0.47420268	0.05718272	8.293	< 0.0000000000000002 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7556 on 20564 degrees of freedom

(7951 observations deleted due to missingness)

Multiple R-squared: 0.2571, Adjusted R-squared: 0.2565

F-statistic: 474.4 on 15 and 20564 DF, p-value: < 0.00000000000000022

```
reg2a <- coeftest(reg2, vcov = vcovCL, type = "HCO", cluster = ~personid)

export_summs(
  reg1b, reg2a,
  model.names = c("Education", "+ASVAB"),
  digits = 4,
```

```

    coefs = c("highested", "asvab", "age", "agesq", "agecu", "union", "tenure", "female")
)

```

LOADING AND CLEANING FIELD OF STUDY

Adding the college majors

```

majors <- read.table(
  "C:/Users/nitya/OneDrive/Desktop/ECO 340 FINAL PROJECT/eco340datapromajorssmaller.dat",
  header = FALSE, sep = " "
)
names(majors)

```

```

[1] "V1" "V2" "V3" "V4" "V5" "V6" "V7" "V8" "V9" "V10" "V11" "V12"
[13] "V13" "V14" "V15" "V16" "V17" "V18" "V19" "V20" "V21" "V22" "V23"

```

Loading names of the data in the NLSY97 as is to identify

```

colnames(majors) <- c(
  "personid",
  "major2013_01", "major2013_02", "major2013_03", "major2013_04",
  "major2015_01", "major2015_02", "major2015_03", "major2015_04",
  "major2017_01", "major2017_02", "major2017_03", "major2017_04", "major2017_05",
  "major2019_01", "major2019_02", "major2019_03", "major2019_04", "major2019_05",

```

```

"major2021_01",
"major2023_01", "major2023_02", "major2023_03"
)

```

Extracting data

```

majors <- majors %>% mutate(across(everything(), as.numeric))

majors_clean <- majors %>%

  mutate(across(-personid, ~ case_when(

    .x < 0 ~ NA_real_,

    .x == 999 ~ NA_real_,

    TRUE ~ .x

  )))

majors_clean <- majors_clean %>%

  mutate(

    major_raw = coalesce(

      major2013_01, major2013_02, major2013_03, major2013_04,

      major2015_01, major2015_02, major2015_03, major2015_04,

      major2017_01, major2017_02, major2017_03, major2017_04, major2017_05,

      major2019_01, major2019_02, major2019_03, major2019_04, major2019_05,

      major2021_01,

      major2023_01, major2023_02, major2023_03
    )
  )

```

```

    )
  )

majors_final <- majors_clean %>% select(personid, major_raw)

sum(!is.na(majors_final$major_raw))

```

```
[1] 1681
```

```
table(majors_final$major_raw, useNA = "ifany")
```

1	3	4	5	9	10	11	12	13	14	15	16	19	22	23	24
5	5	8	3	26	7	80	24	112	44	27	5	24	24	14	83
25	26	27	29	30	31	34	35	36	38	39	40	41	42	43	44
2	25	6	3	21	7	3	2	1	3	10	14	13	59	99	39
45	46	47	48	49	50	51	52	54	60	61	<NA>				
39	8	16	7	10	37	406	338	16	1	5	7303				

Mapping the numbers into field of study groups

```

majors_final <- majors_final %>%

mutate(

  field_group = case_when(

    major_raw %in% c(11, 14, 15, 26, 27, 40, 41) ~ "STEM",

```

```

major_raw %in% c(52) ~ "Business",

major_raw %in% c(42, 43, 44, 45) ~ "Social Science/Public Service",

major_raw %in% c(16, 23, 38, 39, 50, 54, 5, 9) ~ "Humanities/Arts",

major_raw %in% c(13, 32, 33) ~ "Education",

major_raw %in% c(34, 51, 60) ~ "Health",

major_raw %in% c(10, 12, 46, 47, 48, 49) ~ "Trades/Applied",

major_raw %in% c(1, 3, 4, 19, 24, 25, 28, 29, 30, 31, 35, 36, 37, 53, 96) ~ "Other",

TRUE ~ NA_character_

)

)

table(majors_final$field_group, useNA = "ifany")

```

Business	Education
338	112
Health	Humanities/Arts
410	114
Other	Social Science/Public Service
161	236
STEM	Trades/Applied
209	72
<NA>	
7332	

Merging the majors and field of study part into the nlsy agerestric

```
nlsy_agerestric <- nlsy_agerestric %>% mutate(personid = as.numeric(personid))
majors_final <- majors_final %>% mutate(personid = as.numeric(personid))

nlsy_final <- nlsy_agerestric %>% left_join(majors_final, by = "personid")

nrow(nlsy_final)
```

```
[1] 28531
```

```
sum(!is.na(nlsy_final$field_group))
```

```
[1] 4897
```

CREATING A DEGREE VARIABLE

```
degree_max <- nlsy_final %>%

  mutate(highested3 = as.numeric(highested3)) %>%

  filter(highested3 >= 0) %>%

  group_by(personid) %>%

  summarise(degree_raw = max(highested3, na.rm = TRUE)) %>%

  ungroup()
```

```

nlsy_final <- nlsy_final %>%

  select(-any_of("degree_raw")) %>%

  left_join(degree_max, by = "personid")

nlsy_final <- nlsy_final %>%

  mutate(degree_type = case_when(

    degree_raw %in% c(0, 1) ~ "Less than HS / GED",

    degree_raw == 2       ~ "High School Diploma",

    degree_raw == 3       ~ "Associate",

    degree_raw == 4       ~ "Bachelor's",

    degree_raw == 5       ~ "Master's",

    degree_raw == 6       ~ "PhD",

    degree_raw == 7       ~ "Professional",

    TRUE                  ~ NA_character_

  ))

table(nlsy_final$degree_type, useNA = "ifany")

```

Associate	Bachelor's	High School Diploma	Less than HS / GED
2608	6269	10970	4765
Master's	PhD	Professional	<NA>
3077	295	521	26

CREATING A POST SECONDARY EDUCATION DISTINCTION VARIABLE

```
nlsy_dergee <- nlsy_final %>%  
  filter(  
    !is.na(field_group),  
    degree_type %in% c("Associate", "Bachelor's", "Master's", "PhD", "Professional")  
  )  
  
nrow(nlsy_dergee)
```

```
[1] 2895
```

```
length(unique(nlsy_dergee$personid))
```

```
[1] 828
```

```
table(nlsy_dergee$field_group)
```

Business	Education
570	230
Health	Humanities/Arts
780	214
Other Social Science/Public Service	

296	394
STEM	Trades/Applied
317	94

```
table(nlsy_dergee$degree_type)
```

Associate	Bachelor's	Master's	PhD	Professional
1129	1292	405	19	50

FIG 1: MEAN LOG WAGE BY LEVEL OF DEGREE ACHIEVED

```
deg_means <- nlsy_final %>%
  filter(!is.na(degree_type)) %>%
  group_by(degree_type) %>%
  summarise(mean_lnwage = mean(lnwage, na.rm = TRUE), n = n()) %>%
  mutate(degree_type = factor(degree_type,
    levels = c("Less than HS / GED", "High School Diploma", "Associate",
      "Bachelor's", "Master's", "PhD", "Professional")))

fig1 <- ggplot(deg_means, aes(x = degree_type, y = mean_lnwage, fill = degree_type)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  scale_fill_brewer(palette = "Blues") +
  xlab("Highest degree earned") +
```

```

ylab("Mean log hourly wage") +

ggtitle("Figure 1: Mean Log Wage by Degree Level (base)") +

theme_bw() +

theme(axis.text.x = element_text(angle = 30, hjust = 1))

print(fig1)

```

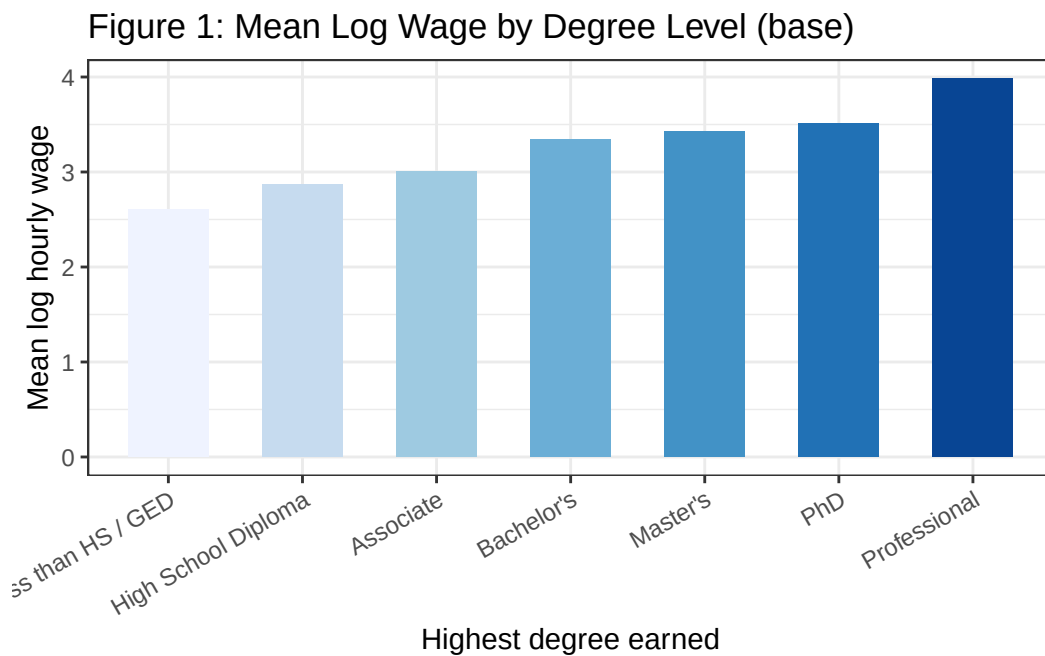


FIG 2: MEAN LOG WAGE BY TYPE OF STUDY

```

field_means <- nlsy_dergee %>%

  group_by(field_group) %>%

  summarise(mean_lnwage = mean(lnwage, na.rm = TRUE), n = n()) %>%

  arrange(mean_lnwage)

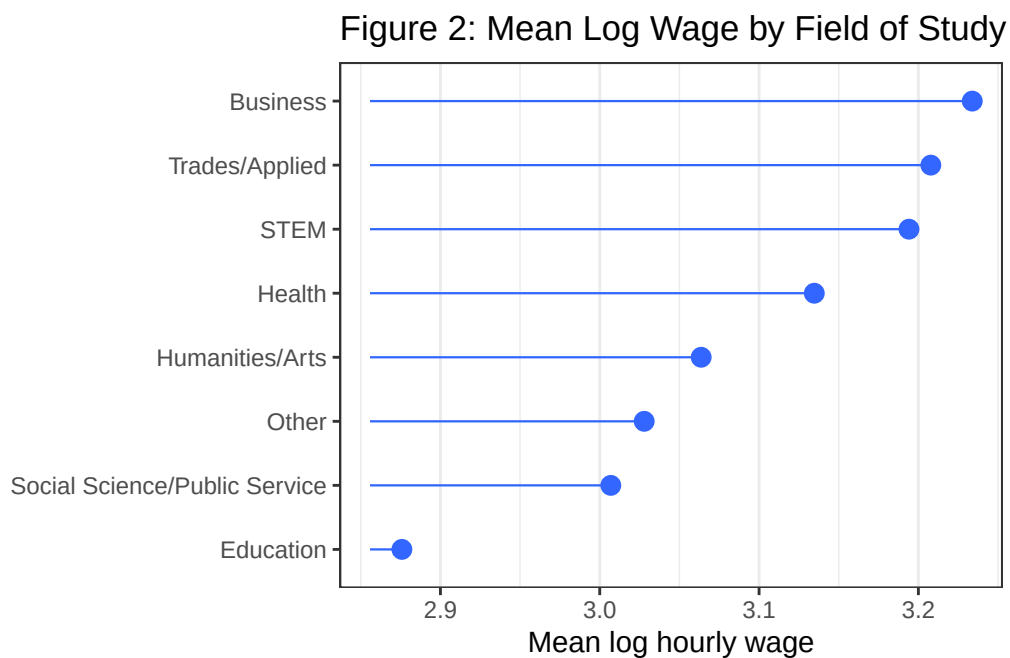
```

```

fig2 <- ggplot(field_means,
               aes(x = mean_lnwage,
                   y = fct_reorder(field_group, mean_lnwage))) +
  geom_point(color = "#3366FF", size = 3) +
  geom_segment(aes(xend = min(mean_lnwage) - 0.02, yend = field_group),
               color = "#3366FF", linewidth = 0.4) +
  xlab("Mean log hourly wage") +
  ylab("") +
  ggtitle("Figure 2: Mean Log Wage by Field of Study (base)") +
  theme_bw() +
  theme(panel.grid.major.y = element_blank())

print(fig2)

```



REGRESSION 3: WAGE BY LEVEL OF DEGREE ACHIEVED

```
nlsy_final$degree_type <- relevel(factor(nlsy_final$degree_type), ref = "High School Diploma")

reg3 <- lm(
  lnwage ~ degree_type + age + agesq + agecu + union + tenure + female + as.factor(year),
  data = nlsy_final
)

summary(reg3)
```

Call:

```
lm(formula = lnwage ~ degree_type + age + agesq + agecu + union +
    tenure + female + as.factor(year), data = nlsy_final)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.4690	-0.3478	0.0217	0.3768	6.8746

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	-6.5616123	6.2878657	-1.044
degree_typeAssociate	0.1835726	0.0178826	10.265

degree_typeBachelor's	0.4877690	0.0129207	37.751
degree_typeLess than HS / GED	-0.2494304	0.0146779	-16.994
degree_typeMaster's	0.5810788	0.0166560	34.887
degree_typePhD	0.6998038	0.0480402	14.567
degree_typeProfessional	1.1716877	0.0362891	32.288
age	0.6999399	0.5252806	1.333
agesq	-0.0175927	0.0145081	-1.213
agecu	0.0001474	0.0001327	1.111
union	0.0403441	0.0134434	3.001
tenure	0.0249044	0.0010317	24.139
female	-0.2272705	0.0098573	-23.056
as.factor(year)2011	-0.0651939	0.0345171	-1.889
as.factor(year)2013	-0.0737066	0.0344563	-2.139
as.factor(year)2015	-0.0528039	0.0388261	-1.360
as.factor(year)2017	0.0971540	0.0435158	2.233
as.factor(year)2019	0.0926219	0.0463947	1.996
as.factor(year)2021	0.1677299	0.0493628	3.398
as.factor(year)2023	0.4707172	0.0523940	8.984

Pr(>|t|)

(Intercept)	0.29671
degree_typeAssociate	< 0.0000000000000002 ***
degree_typeBachelor's	< 0.0000000000000002 ***
degree_typeLess than HS / GED	< 0.0000000000000002 ***
degree_typeMaster's	< 0.0000000000000002 ***

degree_typePhD	< 0.0000000000000002 ***
degree_typeProfessional	< 0.0000000000000002 ***
age	0.18271
agesq	0.22529
agecu	0.26670
union	0.00269 **
tenure	< 0.0000000000000002 ***
female	< 0.0000000000000002 ***
as.factor(year)2011	0.05894 .
as.factor(year)2013	0.03243 *
as.factor(year)2015	0.17384
as.factor(year)2017	0.02558 *
as.factor(year)2019	0.04590 *
as.factor(year)2021	0.00068 ***
as.factor(year)2023	< 0.0000000000000002 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7677 on 24971 degrees of freedom

(3540 observations deleted due to missingness)

Multiple R-squared: 0.2385, Adjusted R-squared: 0.238

F-statistic: 411.7 on 19 and 24971 DF, p-value: < 0.00000000000000022

```
reg3a <- coeftest(reg3, vcov = vcovCL, type = "HCO", cluster = ~personid)
```

REGRESSION 4: WAGE BY LEVEL OF DEGREE ACHIEVED WITH ASVAB AS AN ABILITY CONTROL

```
reg4 <- lm(  
  lnwage ~ degree_type + asvab + age + agesq + agecu + union + tenure + female + as.factor(year)  
  data = nlsy_final  
)  
  
reg4a <- coeftest(reg4, vcov = vcovCL, type = "HCO", cluster = ~personid)  
  
summary(reg4)
```

Call:

```
lm(formula = lnwage ~ degree_type + asvab + age + agesq + agecu +  
    union + tenure + female + as.factor(year), data = nlsy_final)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.5350	-0.3408	0.0207	0.3676	6.7404

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	-2.63147593	6.77268118	-0.389
degree_typeAssociate	0.14000606	0.01903780	7.354
degree_typeBachelor's	0.35553971	0.01484368	23.952
degree_typeLess than HS / GED	-0.14412861	0.01692392	-8.516
degree_typeMaster's	0.40243860	0.01879665	21.410
degree_typePhD	0.47451567	0.05166458	9.185
degree_typeProfessional	0.89822230	0.03889268	23.095
asvab	0.00582047	0.00021840	26.651
age	0.37401793	0.56579765	0.661
agesq	-0.00926011	0.01562801	-0.593
agecu	0.00007757	0.00014293	0.543
union	0.07389341	0.01469829	5.027
tenure	0.02426539	0.00111435	21.775
female	-0.22720463	0.01064152	-21.351
as.factor(year)2011	-0.06379201	0.03777196	-1.689
as.factor(year)2013	-0.05942965	0.03761582	-1.580
as.factor(year)2015	-0.01958132	0.04228708	-0.463
as.factor(year)2017	0.12778458	0.04728692	2.702
as.factor(year)2019	0.11583348	0.05037753	2.299
as.factor(year)2021	0.18904070	0.05352465	3.532
as.factor(year)2023	0.48588679	0.05672885	8.565

Pr(>|t|)

(Intercept)	0.697618	
degree_typeAssociate	0.000000000000199	***
degree_typeBachelor's	< 0.0000000000000002	***
degree_typeLess than HS / GED	< 0.0000000000000002	***
degree_typeMaster's	< 0.0000000000000002	***
degree_typePhD	< 0.0000000000000002	***
degree_typeProfessional	< 0.0000000000000002	***
asvab	< 0.0000000000000002	***
age	0.508591	
agesq	0.553501	
agecu	0.587353	
union	0.000000501489990	***
tenure	< 0.0000000000000002	***
female	< 0.0000000000000002	***
as.factor(year)2011	0.091259	.
as.factor(year)2013	0.114143	
as.factor(year)2015	0.643329	
as.factor(year)2017	0.006891	**
as.factor(year)2019	0.021497	*
as.factor(year)2021	0.000414	***
as.factor(year)2023	< 0.0000000000000002	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7493 on 20550 degrees of freedom

(7960 observations deleted due to missingness)

Multiple R-squared: 0.2683, Adjusted R-squared: 0.2676

F-statistic: 376.8 on 20 and 20550 DF, p-value: < 0.00000000000000022

Table: Degrees with and without ASVAB

```
export_summs(  
  reg3a, reg4a,  
  model.names = c("Degree (no ASVAB)", "Degree + ASVAB"),  
  digits = 4,  
  coefs = c("degree_typeAssociate", "degree_typeBachelor's",  
            "degree_typeMaster's", "degree_typePhD",  
            "degree_typeProfessional", "degree_typeLess than HS / GED",  
            "asvab", "age", "agesq", "agecu", "union", "tenure", "female")  
)
```

REGRESSION 5: WAGES BY FIELD OF STUDY

```
nlsy_dergee$field_group <- relevel(factor(nlsy_dergee$field_group),  
                                     ref = "Social Science/Public Service")  
  
reg5 <- lm(  
  #
```

```

lnwage ~ field_group + age + agesq + agecu + union + tenure + female + as.factor(year),
data = nlsy_dergee
)

summary(reg5)

```

Call:

```

lm(formula = lnwage ~ field_group + age + agesq + agecu + union +
    tenure + female + as.factor(year), data = nlsy_dergee)

```

Residuals:

Min	1Q	Median	3Q	Max
-4.0966	-0.3727	-0.0029	0.3369	6.9539

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-27.3589799	18.9069233	-1.447	0.1480
field_groupBusiness	0.2758376	0.0501407	5.501	0.0000000415280638
field_groupEducation	-0.0893263	0.0646592	-1.381	0.1673
field_groupHealth	0.2435679	0.0485429	5.018	0.0000005599209564
field_groupHumanities/Arts	0.0779741	0.0662996	1.176	0.2397
field_groupOther	0.1033999	0.0592545	1.745	0.0811
field_groupSTEM	0.1478204	0.0586386	2.521	0.0118

field_groupTrades/Applied	0.1396530	0.0927397	1.506	0.1322
age	2.3052456	1.5712604	1.467	0.1425
agesq	-0.0578857	0.0431735	-1.341	0.1801
agecu	0.0004814	0.0003930	1.225	0.2206
union	0.1867754	0.0379270	4.925	0.0000009003060712
tenure	0.0265758	0.0034354	7.736	0.00000000000000148
female	-0.1702619	0.0328140	-5.189	0.0000002287294982
as.factor(year)2011	-0.2027258	0.1277619	-1.587	0.1127
as.factor(year)2013	-0.2626677	0.1187001	-2.213	0.0270
as.factor(year)2015	-0.3625355	0.1333479	-2.719	0.0066
as.factor(year)2017	-0.2055387	0.1480545	-1.388	0.1652
as.factor(year)2019	-0.1646470	0.1563424	-1.053	0.2924
as.factor(year)2021	-0.0906022	0.1632384	-0.555	0.5789
as.factor(year)2023	0.2743755	0.1698177	1.616	0.1063

(Intercept)

field_groupBusiness ***

field_groupEducation

field_groupHealth ***

field_groupHumanities/Arts

field_groupOther .

field_groupSTEM *

field_groupTrades/Applied

age

```

agesq
agecu

union          ***

tenure         ***

female        ***

as.factor(year)2011
as.factor(year)2013      *
as.factor(year)2015      **
as.factor(year)2017
as.factor(year)2019
as.factor(year)2021
as.factor(year)2023

---

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.719 on 2507 degrees of freedom

(367 observations deleted due to missingness)

Multiple R-squared:  0.179, Adjusted R-squared:  0.1725

F-statistic: 27.34 on 20 and 2507 DF,  p-value: < 0.00000000000000022

```

```
reg5a <- coeftest(reg5, vcov = vcovCL, type = "HCO", cluster = ~personid)
```

REGRESSION 6: WAGES BY FIELD OF STUDY + DEGREE

ACHIEVED

```
nlsy_dergee$degree_type <- relevel(factor(nlsy_dergee$degree_type), ref = "Bachelor's")

reg6 <- lm(
  lnwage ~ field_group + degree_type + age + agesq + agecu + union + tenure + female + as.factor(year),
  data = nlsy_dergee
)

summary(reg6)
```

Call:

```
lm(formula = lnwage ~ field_group + degree_type + age + agesq +
    agecu + union + tenure + female + as.factor(year), data = nlsy_dergee)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.9171	-0.3339	0.0058	0.3227	6.9004

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-25.3488708	18.2885134	-1.386	0.165854

field_groupBusiness	0.2725451	0.0485890	5.609	0.00000002256069
field_groupEducation	-0.1841548	0.0635139	-2.899	0.003771
field_groupHealth	0.2758728	0.0475057	5.807	0.00000000715908
field_groupHumanities/Arts	0.0403008	0.0643841	0.626	0.531410
field_groupOther	0.1481344	0.0574574	2.578	0.009989
field_groupSTEM	0.2080494	0.0569055	3.656	0.000261
field_groupTrades/Applied	0.2446574	0.0901097	2.715	0.006671
degree_typeAssociate	-0.2651009	0.0312392	-8.486	< 0.0000000000000002
degree_typeMaster's	0.1727993	0.0435399	3.969	0.00007427984496
degree_typePhD	0.4636336	0.1770910	2.618	0.008897
degree_typeProfessional	0.6573011	0.1075306	6.113	0.00000000113300
age	2.1302095	1.5198949	1.402	0.161173
agesq	-0.0527697	0.0417631	-1.264	0.206510
agecu	0.0004329	0.0003801	1.139	0.254950
union	0.1643169	0.0368913	4.454	0.00000879528282
tenure	0.0245256	0.0033275	7.371	0.000000000000023
female	-0.1724783	0.0318239	-5.420	0.00000006537465
as.factor(year)2011	-0.1869017	0.1236114	-1.512	0.130657
as.factor(year)2013	-0.2373175	0.1148449	-2.066	0.038892
as.factor(year)2015	-0.3371545	0.1290439	-2.613	0.009036
as.factor(year)2017	-0.2005588	0.1432274	-1.400	0.161553
as.factor(year)2019	-0.1706359	0.1512387	-1.128	0.259320
as.factor(year)2021	-0.1161880	0.1579219	-0.736	0.461964
as.factor(year)2023	0.2431578	0.1643336	1.480	0.139090

(Intercept)	
field_groupBusiness	***
field_groupEducation	**
field_groupHealth	***
field_groupHumanities/Arts	
field_groupOther	**
field_groupSTEM	***
field_groupTrades/Applied	**
degree_typeAssociate	***
degree_typeMaster's	***
degree_typePhD	**
degree_typeProfessional	***
age	
agesq	
agecu	
union	***
tenure	***
female	***
as.factor(year)2011	
as.factor(year)2013	*
as.factor(year)2015	**
as.factor(year)2017	
as.factor(year)2019	

```

as.factor(year)2021
as.factor(year)2023
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6954 on 2503 degrees of freedom

(367 observations deleted due to missingness)

Multiple R-squared:  0.2333,    Adjusted R-squared:  0.226

F-statistic: 31.74 on 24 and 2503 DF,  p-value: < 0.00000000000000022

```

```
reg6a <- coeftest(reg6, vcov = vcovCL, type = "HCO", cluster = ~personid)
```

Table: FIELD OF STUDY WITH AND WITHOUT DEGREE AS A CONTROL

```

export_summs(
  reg5a, reg6a,
  model.names = c("Field only", "Field + Degree"),
  digits = 4,
  coefs = c("field_groupSTEM", "field_groupBusiness",
            "field_groupHealth", "field_groupHumanities/Arts",
            "field_groupEducation", "field_groupTrades/Applied", "field_groupOther",
            "degree_typeAssociate", "degree_typeMaster's",
            "degree_typePhD", "degree_typeProfessional",
            "union", "tenure", "female")
)

```

REGRESSION 7: WITH ASVAB AS AN ABILITY CONTROL

```
reg7 <- lm(
  lnwage ~ field_group + degree_type + asvab + age + agesq + agecu + union + tenure + female +
  data = nlsy_dergee
)

summary(reg7)
```

Call:

```
lm(formula = lnwage ~ field_group + degree_type + asvab + age +
    agesq + agecu + union + tenure + female + as.factor(year),
    data = nlsy_dergee)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.6790	-0.3408	0.0076	0.3252	6.7330

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-17.8723292	20.1548618	-0.887	0.37532
field_groupBusiness	0.2742692	0.0543632	5.045	0.00000049262849 ***
field_groupEducation	-0.1957263	0.0695954	-2.812	0.00496 **

field_groupHealth	0.2660977	0.0531469	5.007	0.00000060001015	***
field_groupHumanities/Arts	-0.0118262	0.0720262	-0.164	0.86960	
field_groupOther	0.0821494	0.0646177	1.271	0.20376	
field_groupSTEM	0.1694410	0.0638662	2.653	0.00804	**
field_groupTrades/Applied	0.2522496	0.0951171	2.652	0.00806	**
degree_typeAssociate	-0.2334707	0.0352299	-6.627	0.00000000004351	***
degree_typeMaster's	0.1203592	0.0479795	2.509	0.01220	*
degree_typePhD	0.3574837	0.1782746	2.005	0.04507	*
degree_typeProfessional	0.5100341	0.1130260	4.513	0.00000676376863	***
asvab	0.0035665	0.0006170	5.780	0.00000000858870	***
age	1.4855446	1.6745872	0.887	0.37512	
agesq	-0.0347639	0.0460041	-0.756	0.44993	
agecu	0.0002683	0.0004187	0.641	0.52168	
union	0.1800151	0.0412036	4.369	0.00001310172812	***
tenure	0.0265714	0.0037304	7.123	0.00000000000145	***
female	-0.1830364	0.0356596	-5.133	0.00000031193190	***
as.factor(year)2011	-0.2520467	0.1337007	-1.885	0.05955	.
as.factor(year)2013	-0.2978557	0.1256247	-2.371	0.01783	*
as.factor(year)2015	-0.4047040	0.1425355	-2.839	0.00456	**
as.factor(year)2017	-0.2924362	0.1577184	-1.854	0.06386	.
as.factor(year)2019	-0.3017916	0.1668813	-1.808	0.07069	.
as.factor(year)2021	-0.2455624	0.1740965	-1.410	0.15854	
as.factor(year)2023	0.1142028	0.1812162	0.630	0.52863	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6954 on 2076 degrees of freedom

(793 observations deleted due to missingness)

Multiple R-squared: 0.2481, Adjusted R-squared: 0.2391

F-statistic: 27.4 on 25 and 2076 DF, p-value: < 0.00000000000000022

```
reg7a <- coeftest(reg7, vcov = vcovCL, type = "HCO", cluster = ~personid)
```

TABLE: FIELD AND DEGREES BEFORE AND AFTER ADDING ASVAB AS A CONTROL

```
export_summs(  
  reg6a, reg7a,  
  model.names = c("Field + Degree (no ASVAB)", "Field + Degree + ASVAB"),  
  digits = 4,  
  coefs = c("field_groupSTEM", "field_groupBusiness",  
            "field_groupHealth", "field_groupHumanities/Arts",  
            "field_groupEducation", "field_groupTrades/Applied", "field_groupOther",  
            "asvab", "union", "tenure", "female")  
)
```

Figure 4: Coefficient plot — field premiums with 95% CI

```
field_coefs <- tidy(reg6a) %>%  
  
  filter(grepl("field_group", term)) %>%  
  
  mutate(  
  
    field = gsub("field_group", "", term),  
  
    lower = estimate - 1.96 * std.error,  
  
    upper = estimate + 1.96 * std.error  
  
  )  
  
fig4 <- ggplot(field_coefs,  
  
               aes(x = estimate,  
  
                   y = fct_reorder(field, estimate),  
  
                   color = estimate > 0)) +  
  
  geom_point(size = 3) +  
  
  geom_errorbarh(aes(xmin = lower, xmax = upper), height = 0.3) +  
  
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey50") +  
  
  scale_color_manual(values = c("TRUE" = "#3366FF", "FALSE" = "#CC3300"),  
  
                     guide = "none") +  
  
  xlab("Coefficient (ref: Social Science/Public Service)") +  
  
  ylab("") +  
  
  ggtitle("Figure 4: Field of Study Wage Premiums\n(95% CI, conditional on degree, age, control variables)") +  
  
  theme_bw()
```

```
print(fig4)
```

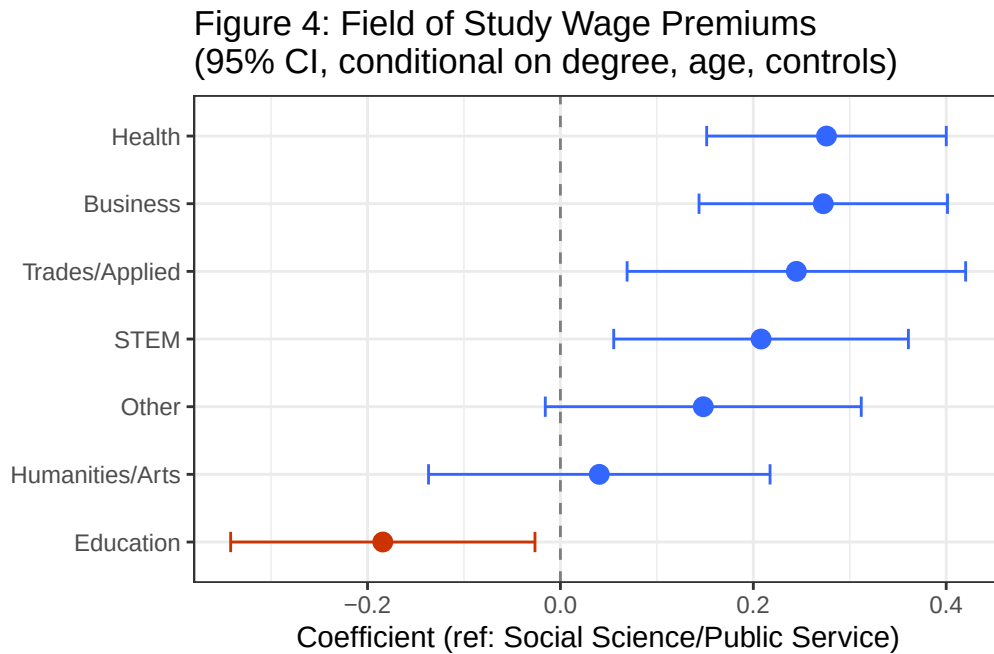


Figure 5: STEM and Health premium — with vs. without ASVAB

```
stems <- data.frame(
  field = rep(c("STEM", "Health"), 2),
  model = c("No ASVAB", "No ASVAB", "With ASVAB", "With ASVAB"),
  est   = c(coef(reg6)["field_groupSTEM"],
            coef(reg6)["field_groupHealth"],
            coef(reg7)["field_groupSTEM"],
            coef(reg7)["field_groupHealth"]),
  se    = c(sqrt(vcov(reg6)["field_groupSTEM", "field_groupSTEM"]),
            sqrt(vcov(reg6)["field_groupHealth", "field_groupHealth"])),
```

```

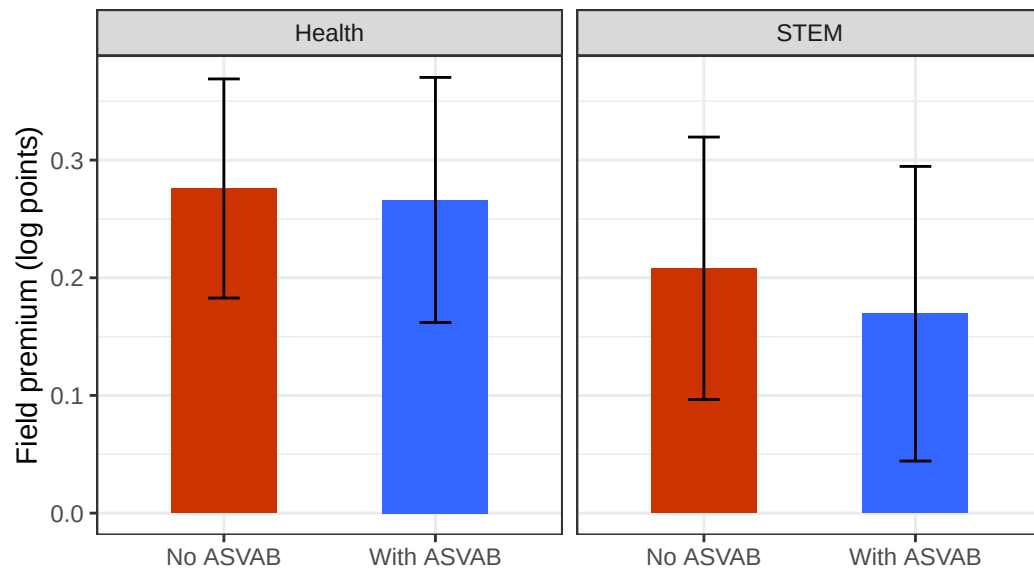
      sqrt(vcov(reg7)["field_groupSTEM", "field_groupSTEM"]),
      sqrt(vcov(reg7)["field_groupHealth", "field_groupHealth"]))
)

fig5 <- ggplot(stems, aes(x = model, y = est, fill = model)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_errorbar(aes(ymin = est - 1.96 * se, ymax = est + 1.96 * se), width = 0.15) +
  facet_wrap(~field) +
  scale_fill_manual(values = c("No ASVAB" = "#CC3300", "With ASVAB" = "#3366FF")) +
  xlab("") +
  ylab("Field premium (log points)") +
  ggtitle("Figure 5: STEM and Health Premiums Before and After Controlling for ASVAB") +
  theme_bw()

print(fig5)

```

Figure 5: STEM and Health Premiums Before and After Contro



year	n
1.99e+03	8984
2e+03	8984
2e+03	8984
2e+03	8984
2e+03	8984
2e+03	8984
2e+03	8984
2e+03	8984
2e+03	8984
2e+03	8984
2e+03	8984
2.01e+03	8984
2.01e+03	8984
2.01e+03	8984
2.01e+03	8984
2.01e+03	8984
2.01e+03	8984
2.01e+03	8984
2.01e+03	8984
2.01e+03	8984
2.01e+03	8984
2.02e+03	8984
2.02e+03	8984
2.02e+03	8984
2.02e+03	8984
2.02e+03	8984
2.02e+03	8984
2.02e+03	8984
2.02e+03	8984
2.02e+03	8984

year	n
2e+03	8975
2e+03	8982
2e+03	8982
2e+03	8982
2e+03	8982
2e+03	8982
2e+03	8982
2e+03	8982
2e+03	8982
2.01e+03	8982
2.01e+03	8982
2.01e+03	8982
2.01e+03	8982
2.01e+03	8982
2.01e+03	8982
2.01e+03	8982
2.01e+03	8982
2.02e+03	8982
2.02e+03	8982
2.02e+03	8982
2.02e+03	8982

age	highested
13	3
14	4
15	4
16	1
17	2
18	1
19	1
20	1
21	1
22	1
23	1
24	1
25	1
26	1
27	1
28	1
29	1
30	1
31	2
32	1
33	2
34	1
35	2
36	1
37	2
38	1
39	2
40	1
41	2
42	3 ₄₉
43	2

	Basic	Robust SE	Clustered SE
highested	0.1015 *** (0.0017)	0.1015 *** (0.0017)	0.1015 *** (0.0026)
age	0.6268 (0.5294)	0.6268 (0.5256)	0.6268 (0.4790)
agesq	-0.0157 (0.0146)	-0.0157 (0.0145)	-0.0157 (0.0132)
agecu	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)
union	0.0254 (0.0135)	0.0254 (0.0134)	0.0254 (0.0184)
tenure	0.0256 *** (0.0010)	0.0256 *** (0.0010)	0.0256 *** (0.0015)
female	-0.2354 *** (0.0099)	-0.2354 *** (0.0099)	-0.2354 *** (0.0151)
N	25010	25010	25010
R2	0.2262		

*** p < 0.001; ** p < 0.01; * p < 0.05.

	Education	+ASVAB
highested	0.1015 *** (0.0026)	0.0665 *** (0.0032)
asvab		0.0063 *** (0.0003)
age	0.6268 (0.4790)	0.3554 (0.5185)
agesq	-0.0157 (0.0132)	-0.0088 (0.0143)
agecu	0.0001 (0.0001)	0.0001 (0.0001)
union	0.0254 (0.0184)	0.0650 ** (0.0204)
tenure	0.0256 *** (0.0015)	0.0244 *** (0.0016)
female	-0.2354 *** (0.0151)	-0.2312 *** (0.0162)
nobs	25010	20580
logLik	-29082.2790	-23425.9920
AIC	58196.5600	46885.9800
BIC	58326.5900	47020.8300
nobs.1	25010.0000	20580.0000

*** p < 0.001; ** p < 0.01; * p < 0.05.

	Degree (no ASVAB)	Degree + ASVAB
degree_typeAssociate	0.1836 *** (0.0255)	0.1400 *** (0.0277)
degree_typeBachelor's	0.4878 *** (0.0199)	0.3555 *** (0.0224)
degree_typeMaster's	0.5811 *** (0.0234)	0.4024 *** (0.0265)
degree_typePhD	0.6998 *** (0.0529)	0.4745 *** (0.0587)
degree_typeProfessional	1.1717 *** (0.0492)	0.8982 *** (0.0543)
degree_typeLess than HS / GED	-0.2494 *** (0.0234)	-0.1441 *** (0.0256)
asvab		0.0058 *** (0.0003)
age	0.6999 (0.4774)	0.3740 (0.5162)
agesq	-0.0176 (0.0132)	-0.0093 (0.0142)
agecu	0.0001 (0.0001)	0.0001 (0.0001)
union	0.0403 * (0.0184)	0.0739 *** (0.0203)
tenure	0.0249 *** (0.0015)	0.0243 *** (0.0016)
female	-0.2273 *** (0.0149)	-0.2272 *** (0.0159)
nobs	24991	20571
logLik	-28845.1930	-23242.4160
AIC	57732.3900	46528.8300
BIC	57903.0400	46703.3300
nobs.1	24991.0000	20571.0000

	Field only	Field + Degree
field_groupSTEM	0.1478 (0.0803)	0.2080 ** (0.0780)
field_groupBusiness	0.2758 *** (0.0681)	0.2725 *** (0.0657)
field_groupHealth	0.2436 *** (0.0672)	0.2759 *** (0.0634)
field_groupHumanities/Arts	0.0780 (0.0937)	0.0403 (0.0903)
field_groupEducation	-0.0893 (0.0844)	-0.1842 * (0.0805)
field_groupTrades/Applied	0.1397 (0.0946)	0.2447 ** (0.0895)
field_groupOther	0.1034 (0.0837)	0.1481 (0.0836)
degree_typeAssociate		-0.2651 *** (0.0413)
degree_typeMaster's		0.1728 ** (0.0571)
degree_typePhD		0.4636 ** (0.1428)
degree_typeProfessional		0.6573 *** (0.1187)
union	0.1868 *** (0.0485)	0.1643 *** (0.0478)
tenure	0.0266 *** (0.0048)	0.0245 *** (0.0045)
female	-0.1703 *** (0.0433)	-0.1725 *** (0.0421)
nobs	2528	2528
logLik	-2742.4960	-2656.0580
AIC	5528.9920	5364.1160

	Field + Degree (no ASVAB)	Field + Degree + ASVAB
field_groupSTEM	0.2080 ** (0.0780)	0.1694 (0.0875)
field_groupBusiness	0.2725 *** (0.0657)	0.2743 *** (0.0708)
field_groupHealth	0.2759 *** (0.0634)	0.2661 *** (0.0670)
field_groupHumanities/Arts	0.0403 (0.0903)	-0.0118 (0.1000)
field_groupEducation	-0.1842 * (0.0805)	-0.1957 * (0.0903)
field_groupTrades/Applied	0.2447 ** (0.0895)	0.2522 ** (0.0883)
field_groupOther	0.1481 (0.0836)	0.0821 (0.0872)
asvab		0.0036 *** (0.0009)
union	0.1643 *** (0.0478)	0.1800 *** (0.0538)
tenure	0.0245 *** (0.0045)	0.0266 *** (0.0052)
female	-0.1725 *** (0.0421)	-0.1830 *** (0.0471)
nobs	2528	2102
logLik	-2656.0580	-2206.0270
AIC	5364.1160	4466.0540
BIC	5515.8310	4618.6220
nobs.1	2528.0000	2102.0000

*** p < 0.001; ** p < 0.01; * p < 0.05.